

WeatherGPT — Complete System Documentation

AI Meteorological Intelligence Platform • Full Architecture, File Directory & Execution Manual

1. Executive Summary

WeatherGPT is a state-of-the-art AI-powered meteorological intelligence system designed to bridge the gap between complex scientific atmospheric telemetry and human decision support. Built with **Google Cloud Gemini 1.5 Flash (NLU & Vision)**, **Open-Meteo & IMD Telemetry**, a **13 Indian Languages Native Voice Engine (STT/TTS)**, and **Permanent Home Location Persistence**, WeatherGPT empowers citizens, farmers, disaster managers, and travellers with real-time actionable insights.

2. Complete File Directory & Component Architecture

Below is the exhaustive breakdown of every file included in the repository and its specific technical responsibility:

File Path / Name	Description & Responsibility
README.md	Main documentation guide, slide topics for presentations, local commands & cloud deployment guides.
vercel.json	Root configuration for 1-click cloud deployment on Vercel (Vite framework & SPA rewrite rules).
docker-compose.yml	Container orchestration config for launching frontend (Vite) and backend (FastAPI) together.
.gitignore	Excludes node_modules, dist, __pycache__, .env files from version control.
.env.example	Template defining Google Cloud Gemini API key environment variables.
backend/app/main.py	FastAPI Python backend server. Exposes /api/ai/chat, Open-Meteo proxy routes, CORS, and WebSocket alert streams.
backend/requirements.txt	Python dependencies list (fastapi, uvicorn, httpx, asyncio).
frontend/index.html	HTML5 entryptoint with Google Fonts (Outfit, Inter) and PWA manifest links.
frontend/vite.config.ts	Vite bundler configuration file.
frontend/tailwind.config.js	Tailwind CSS configuration defining saffron (#FF9933), india green (#138808), and glassmorphism design tokens.
frontend/src/main.tsx	React 18 DOM root mounting script.
frontend/src/App.tsx	Central App Router, state management, Light/Dark theme handler, permanent location persistence, and header navbar.
frontend/src/index.css	Global CSS tokens, glass-card, glass-panel utilities, animation keyframes, and Light/Dark overrides.
src/types/index.ts	TypeScript data interfaces for WeatherData, HourlyForecast, VisionResult, ChatMessage, and Language Types.
src/services/aiAssistant.ts	Integrates Google Cloud Gemini 1.5 Flash (NLU), Gemini Vision API, General Science Knowledge Engine, and response parser.
src/services/i18n.ts	Multilingual dictionary supporting 13 Indian languages (en, hi, ta, te, kn, ml, mr, bn, gu, pa, or, as, ur).

src/services/weatherApi.ts	Open-Meteo & IMD API integration + Levenshtein Distance Fuzzy String Matching algorithm for location resolution.
src/components/UI/Navbar.tsx	Header bar with City Search, GPS button, Permanent Home button, Light/Dark toggle, NLP info, and mobile menu drawer.
src/components/UI/LanguageWelcomeModal.tsx	Startup language selection modal ('ask before starting') with native voice test buttons.
src/components/UI/VoiceButton.tsx	Speech Recognition (STT) and Neural Speech Synthesis (TTS) button component.
src/components/UI/CameraModal.tsx	Webcam/phone camera photo capture modal for instant sky vision analysis.
src/components/Weather/WeatherCard.tsx	Live dashboard card displaying temperature, condition, metrics, sunrise/sunset, and Permanent Home badge.
src/components/Weather/ForecastCard.tsx	24-Hour hourly horizontal trend list and 7-day extended forecast card.
src/components/Charts/WeatherChart.tsx	Interactive Recharts meteorological graphs for temperature, precipitation probability, and humidity trends.
src/components/Map/LeafletWeatherMap.tsx	Interactive Leaflet GIS map with Doppler radar echo overlays.
src/components/Vision/ImageUploader.tsx	Weather Vision AI photo diagnostic page with sample photographs and custom upload drop zone.
src/components/Alerts/DisasterAlertBanner.tsx	IMD severe weather alert banner with one-click WhatsApp and SMS emergency broadcast sharing.
src/components/Pages/ChatPage.tsx	WeatherGPT AI Chatbot interface with quick prompt pills, floating mobile input toolbar, and Google Cloud API key modal.
src/components/Pages/FloodRiskPage.tsx	Flood Risk AI Analyzer portal for watershed and surface runoff risk.
src/components/Pages/DisasterIntelPage.tsx	Disaster Operations dashboard for active IMD warnings and emergency geofences.
src/components/Pages/ClimatePage.tsx	Long-term climate anomaly and trend analytics portal.
src/components/Pages/AgriculturePage.tsx	Farm Weather Advisor portal providing crop-specific irrigation, pest risk, and harvest advice.
src/components/Pages/AirQualityPage.tsx	Live Air Quality Index (AQI) meter with PM2.5, PM10, and health advisories.
src/components/Pages/MarineAviationPage.tsx	Marine wave height/tide forecasts and Aviation METAR/TAF reports.
src/components/Pages/LocationsSettingsPage.tsx	Saved monitored locations and alert notification preferences management.
src/components/Pages/AdminDashboardPage.tsx	Administrator portal for pipeline telemetry and system status.

3. How to Run WeatherGPT Locally

Follow these simple steps to run the complete WeatherGPT application on your computer:

Step 1: Prerequisites

Ensure you have **Node.js** (v18 or higher) and **Python** (v3.9 or higher) installed on your system.

Step 2: Run Frontend (React + Vite)

Open a terminal window and execute:

```
cd frontend
npm install
npm run dev
```

The frontend dev server will launch at: **http://localhost:5173/**

Step 3: Run Backend (FastAPI Python Server)

Open a second terminal window and execute:

```
cd backend
pip install -r requirements.txt
python -m uvicorn app.main:app --port 8000 --reload
```

The backend API server will launch at: **http://localhost:8000/**

4. How to Deploy to Production (100% Free)

Option A: Deploy Frontend on Vercel (1-Click Deployment)

1. Go to **Vercel.com** and sign in with GitHub.
2. Click 'Add New' → 'Project' and select **PRANAV-26-mv/weatherapp**.
3. Vercel detects root **vercel.json** automatically! Click **Deploy**.
4. Your site will be live at **https://weatherapp-xxx.vercel.app** in under 60 seconds.

Option B: Deploy FastAPI Backend on Render.com

1. Go to **Render.com** → 'New +' → 'Web Service'.
2. Connect repository **PRANAV-26-mv/weatherapp**.
3. Set Root Directory: **backend** | Environment: **Python 3**
4. Build Command: **pip install -r requirements.txt**
5. Start Command: **uvicorn app.main:app --host 0.0.0.0 --port \$PORT**
6. Click **Create Web Service**. Your API will be live at **https://weatherapp-backend.onrender.com**.

5. Innovation & Feature Highlights

- **Multilingual Voice Engine (13 Indian Languages)**: Prompts user on startup for preferred language. Speech input (STT) and neural voice readout (TTS) run natively in chosen language.
- **Multimodal Weather Vision AI**: Analyzes uploaded cloud, sky, or radar photos for storm risk & safety precautions using Google Cloud Gemini Vision.
- **Permanent Home Location Memory**: Automatically saves selected city (e.g. Sathyamangalam) and reloads it on all future visits.
- **Fuzzy Location Engine**: Employs Levenshtein String Distance matching to correctly resolve town typos (e.g. *sathymagalam* → Sathyamangalam).

- **Mobile Optimized Responsive UI:** Features floating input toolbar, search drawer, and touch-first layout.